

Lab-1.3

Apply the K-medoids clustering on the following data and form the cluster for the value of $K=2$.

Sn.	X	Y
0	8	7
1	3	7
2	4	9
3	9	6
4	8	5
5	9	8
6	7	3
7	8	4
8	7	5
9	4	5

→ Let us randomly select 2 medoids, so select $K=2$ and let $C1-(4,5)$ and $C2-(8,5)$ are the two medoids.

The dissimilarity of each non-medoid point with the medoids is calculated and tabulated:-

$$\text{Manhattan Distance} := |X_o - X_c| + |Y_o - Y_c|$$

Sr.	X	Y	Dissimilarity from C_1	Dissimilarity from C_2
0	8	7	$ 8-4 + 7-5 = 6$	$ 8-7 + 7-5 = 2$
1	3	7	8	7
2	4	9	4	8
3	9	6	6	2
5	5	8	4	6
6	7	3	5	3
7	8	4	5	1
8	7	5	3	1

Each point is assigned to the cluster of that medoid whose dissimilarity is less.

The points 1, 2, 5 go to cluster C_1 and 0, 3, 6, 7, 8 go to cluster C_2 .

$$\text{The cost} = (3+4+4) + (2+2+3+1+1) = 20$$

Randomly select one non-medoid point and recalculate the cost.

Let the randomly selected point be (8, 4).

The dissimilarity of each non-medoid point with the medoids - $C_1(4, 5)$ and $C_2(8, 4)$ is calculated and tabulated:-

Ser.	X	Y	Dissimilarity from G ₁	Dissimilarity from G ₂
0	8	7	6	3
1	3	7	3	8
2	4	9	4	9
3	9	6	6	3
4	8	5	4	1
5	5	8	4	7
6	7	3	5	2
8	7	5	3	2

Each point is assigned to that cluster whose dissimilarity is less. So, the points 1, 2, 5 go to cluster C₁ and 0, 3, 4, 6, 8 go to cluster C₂.

$$\text{The New cost, } (3+4+4) + (3+3+1+2+2) = 22$$

$$\begin{aligned} \text{Swap cost} &= \text{New Cost} - \text{Previous Cost} \\ &= 22 - 20 \\ &= 2 \end{aligned}$$

So, $2 > 0$ that is positive, now our previous medoid is best.

The total cost of medoid (8, 4) > the total cost when (8, 5) was the medoid earlier and it generates the same clusters as earlier.

Hence (4, 5) and (8, 5) are the final medoids.